Q. A point charge $q = 5\,\mu C$ is placed at the center of a cube of side $a = 2\,m$.
Find the electric flux through:

1. The entire cube
2. One face of the cube

Q. An infinite line charge has linear charge density

$$\lambda = 3 \times 10^{-6} C/m$$

Find the electric field at distance $r = 0.2\,m$.

Q. Surface charge density for an uniformly plane sheet of charge is $\sigma = 4 \times 10^{-6} C/m^2$
Find electric field.

Q. An insulator in the form of a solid sphere of radius R has uniform charge density ρ.
Find electric field:

1. Inside ($r < R$)
2. Outside ($r > R$)

$$\rho = \frac{Q}{V}$$



$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0} = \frac{\rho V}{\varepsilon_0}$$

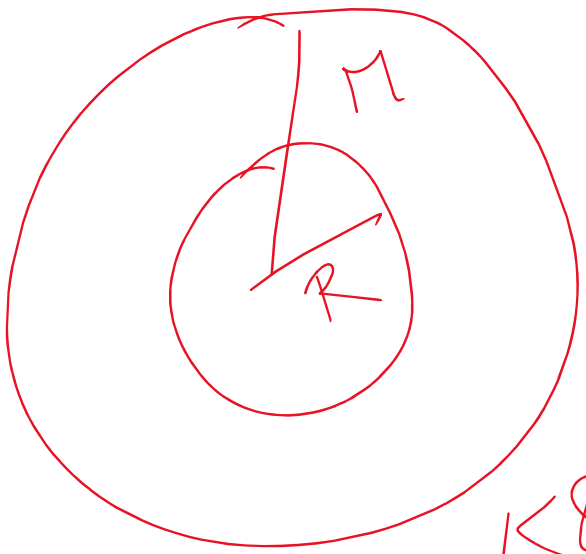
$$E \times 4\pi r^2 = \frac{\rho \left(\frac{4}{3}\pi r^3\right)}{\varepsilon_0}$$

$$E = \frac{\rho r}{3\varepsilon_0}$$

$$E \propto r$$

$E \propto r$ $\quad r < R$ $\quad E = \dfrac{Q}{\frac{4}{3}\pi R^3} \cdot \dfrac{r}{3\varepsilon_0} = \dfrac{Qr}{4\pi\varepsilon_0 R^3}$

$r > R$



$$E\, 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$E = \frac{KQ}{r^2}$$